

# Alex Coker

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## Education

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**University of New Mexico** Expected Spring 2027

*B.S. in Mechanical Engineering | GPA: 3.33/4.0*

- Coursework in dynamics, control systems, and numerical methods

## Research Experience

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**Learning and Control Lab — University of New Mexico** Oct. 2025 – Present

*Undergraduate Research Assistant | Advisor: Prof. Leilei Cui*

- Implemented learned locomotion policies on the Unitree Go2 quadruped, supporting PhD student research on sim-to-real transfer
- Contributed to a whole-body control project targeting dynamic handstand behaviors on the Go2
- Designing a Control Barrier Function (CBF) safety filter for the Go2, enforcing hard safety constraints during agile maneuvers
- Conducted independent study of safety-critical control: CBFs, Hamilton-Jacobi reachability analysis, and Model Predictive Path Integral (MPPI) control
- Running parallel simulation experiments to evaluate and iterate on policies before hardware deployment

## Industry Experience

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**Sandia National Laboratories** June 2025 – Present

*Mechanical Engineering Intern | PV Reliability Group*

- Designing and executing experiments to characterize photovoltaic (PV) connector reliability under thermomechanical loading
- Developing a novel thermomechanical method to measure contact displacement in PV connectors; targeting peer-reviewed publication
- Collecting and analyzing experimental datasets to identify failure modes and inform connector design improvements

## Technical Skills

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**Programming:** Python, C++

**Robotics & Control:** Control Barrier Functions (CBF), Hamilton-Jacobi Reachability, MPPI, Sim-to-Real Transfer, Legged Robot Locomotion (Unitree Go2), Policy Implementation

**Simulation & Tools:** Physics-based simulation (MuJoCo / Isaac Gym), Git, Linux, Windows

**Hardware & Lab:** Hand tools, experimental design, data acquisition, network & computer setup, AI-assisted development (Claude Code)

**In Progress:** ROS (Robot Operating System)

## Honors & Awards

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- 1st Place — UNM Rainforest Innovations Pitch Competition (two-time winner)
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